

Advanced Econometrics

Cointegration

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The Problem: Spurious Regression

A thought experiment

Regress the **number of pirates worldwide** on **global temperature**. You may find a highly “significant” relationship — but it is completely **fake**.

Why does this happen?

- Both series *trend over time* — they share a common drift, not a causal link.
- Standard OLS t -statistics and R^2 are **misleading** for non-stationary data.
- The regression picks up the shared trend, not a genuine relationship.

Spurious Regression

Running OLS on two **unrelated non-stationary** ($I(1)$) variables produces artificially significant results. **Never** interpret such a regression at face value.

Why Does It Go Wrong?

Stationary series

- Fluctuates around a *fixed mean*
- Deviations are temporary
- OLS works as expected

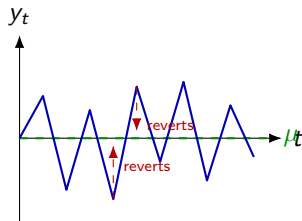
Non-stationary I(1) series

- Has a *stochastic trend* (random walk)
- Deviations are **permanent**
- OLS standard errors are **invalid**
- Two unrelated I(1) series can appear correlated

Key takeaway

Before running any regression on time-series data, **test for stationarity**.

Integration Order: $I(0)$ vs $I(1)$



$I(0)$ — Stationary

- Fluctuates around fixed mean μ
- Deviations are **temporary**
- Shocks die out over time
- OLS works correctly



$I(1)$ — Non-stationary

- **No** fixed mean to return to
- Shocks are **permanent**
- Variance grows without bound
- OLS standard errors are **invalid**

The link between them

y_t is $I(1) \iff \Delta y_t = y_t - y_{t-1}$ is $I(0)$. (“one difference makes it stationary”)

The Drunk Analogy

$I(0)$: drunk on a leash



Staggers everywhere but the leash
always pulls her back

$I(1)$: no leash



Wherever she ends up tonight is
the **new starting point** tomorrow

The leash = mean reversion = stationarity

Two More Tricks

The number = the cure dosage

$I(d) \implies$ difference d times to get $I(0)$

Order	Differences needed	Example
$I(0)$	0 — already stationary	Daily temperature anomaly
$I(1)$	1 — $\Delta y_t = y_t - y_{t-1}$	Stock <i>price</i>
$I(2)$	2 — difference twice	Some macro aggregates

The stock market anchor

Stock price $\sim I(1)$

Today's price = yesterday's
price + random shock.
Shocks are permanent.

Daily return $\sim I(0)$

$r_t = \ln(P_t/P_{t-1})$
Returns fluctuate around zero.
One difference = one return.

Whenever you see **$I(1)$** , think *stock price*. Whenever you see **$I(0)$** , think *stock return*.

Testing for $I(1)$: The ADF Test

We use the **Augmented Dickey–Fuller (ADF)** test.

H_0 : unit root (non-stationary) H_1 : stationary

Testing CPI and PPI from the lab:

```
# Test levels (with constant)
testdf2(variable = ppi_cpi$cpi, test.type = "c",
         max.augmentations = 12, max.order = 12)
# Test first differences (no constant)
testdf2(variable = ppi_cpi$dcpi, test.type = "nc",
         max.augmentations = 12, max.order = 12)
```

Series	ADF on levels	ADF on Δ
CPI	Cannot reject H_0	Reject H_0
PPI	Cannot reject H_0	Reject H_0

$\Rightarrow$ Both CPI and PPI are **$I(1)$** .

ADF Test Types

test.type	Model estimated	Null hypothesis
"nc"	Random walk (no constant)	RW vs stationarity
"c"	Random walk with drift	RW+drift vs stationarity
"ct"	RW around linear trend	RW vs trend-stationarity

Rule of thumb

- Test the **level** with "c" (allows for drift in the data).
- Test the **first difference** with "nc" (differences rarely need a trend).

Cointegration: The Core Idea

Definition

Two $I(1)$ series x_t and y_t are **cointegrated** if there exists a vector $[\beta_0, \beta_1]$ such that

$$y_t - \beta_0 - \beta_1 x_t \sim I(0).$$

The vector $[1, -\beta_0, -\beta_1]$ is called the **cointegrating vector**.

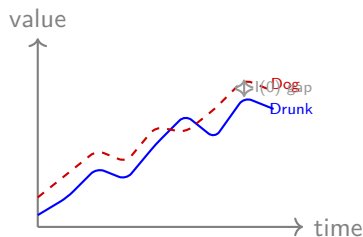
Intuition: Each series wanders on its own, but they are *tied together* in the long run. The gap between them cannot grow forever.

The Drunk and Her Dog

An analogy:

- A **drunk** walks home erratically — $I(1)$.
- Her **dog** also wanders randomly — $I(1)$.
- But the dog *never strays too far* from her.
- The **distance** between them is stationary — $I(0)$.

⇒ Drunk and dog are **cointegrated**.



CPI and PPI: The Cointegrating Relationship

From the lab, the estimated long-run relationship is:

$$\widehat{\text{CPI}}_t = 10.700 + 1.229 \cdot \text{PPI}_t$$

The **cointegrating vector** is $[1, -10.700, -1.229]$, meaning:

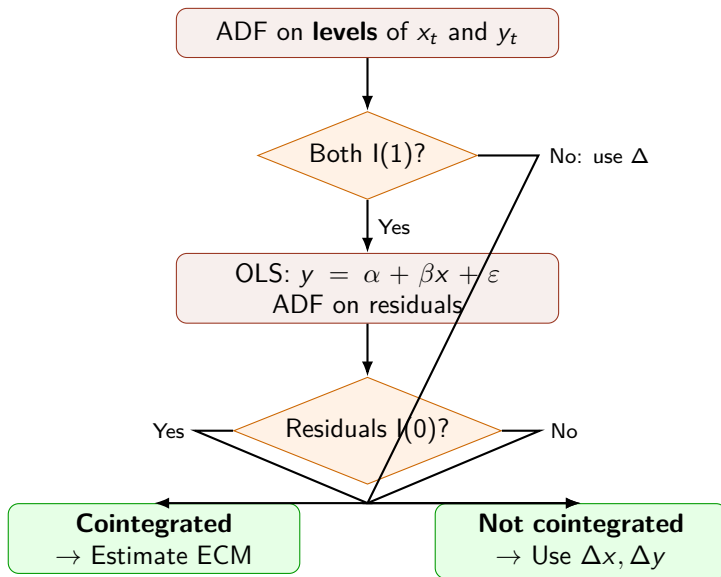
$$\underbrace{\text{CPI}_t - 10.700 - 1.229 \cdot \text{PPI}_t}_{\text{residual } \hat{\varepsilon}_t} \sim I(0)$$

The residual captures the *equilibrium error* — how far the system is from its long-run path at any point in time.

Interpretation

In the long run, a 1-unit increase in PPI is associated with a **1.229-unit increase in CPI**.

Decision Flowchart



Cointegration Test

Step 1: Confirm both are $I(1)$

```
testdf2(variable=mrates$r10y, test.type="c",
        max.augmentations=4, max.order=4)      # levels: cannot reject
        H0
testdf2(variable=diff(mrates$r10y), test.type="nc",
        max.augmentations=4, max.order=4)      # differences: reject H0
# r10y ~ I(1). Repeat for r3m -> r3m ~ I(1)
```

Step 2: Estimate cointegrating regression and test residuals

```
model.coint <- lm(r10y ~ r3m, data=mrates)
testdf2(variable=residuals(model.coint),
        test.type="nc", max.augmentations=4, max.order=4)
```

Result

Test statistic = -2.62

Critical value (5%) ≈ -4.24

$-2.62 > -4.24 \Rightarrow$ **fail to reject H_0**

Residuals are **non-stationary**

$\Rightarrow$ R3M and R10Y are **NOT cointegrated**

Why surprising?

Theory predicts cointegration.

Data say no — at least over this sample.

Possible reasons: structural breaks, near-unit-root behaviour, low power of test.

Summary of Cointegration

- ❶ **Spurious regression** is a real danger — always check integration order before running OLS on time-series data.
- ❷ A series is **I(1)** if its level has a unit root but its first difference is stationary (ADF test).
- ❸ Two I(1) series are **cointegrated** if a linear combination of them is stationary — they share a long-run equilibrium.
- ❹ Test via **Engle–Granger**: OLS regression → ADF on residuals with *special* critical values.
- ❺ Cointegration is the prerequisite for estimating an **Error Correction Model (ECM)**.

What Cointegration Really Measures

*Correlation tells you two things move together **today**.*

*Cointegration tells you they **cannot drift apart forever**.*

What you get

- Long-run **equilibrium** relationship
- Measure of **distance** from that equilibrium
- Speed and direction of **correction**

The key social science question

Is this relationship **real and structural** — or are two things just drifting in the same direction **by coincidence**?

The ECM adjustment coefficient answers:
*when the system is knocked out of equilibrium,
how fast and through what mechanism does it correct?*

Example 1: Public Opinion & Government Policy

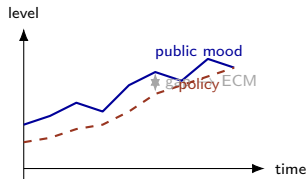
The question

Do governments actually respond to voters in the long run?

Erikson, MacKuen & Stimson (2002)

The Macro Polity

- Tested whether US **public mood** (liberal/conservative index) and **policy outputs** are cointegrated
- **Found:** they are — adjustment takes $\approx 2-3$ years
- Error correction term = **political pressure**



First rigorous evidence that democratic responsiveness is a **real long-run equilibrium**

Example 2: Crime & Incarceration

The question

Does locking people up actually reduce crime — or is the correlation spurious?

The breakdown is itself a finding

Structural instability of the cointegrating vector $\Rightarrow$ other forces dominated (drugs, policing, demographics)

Marvell & Moody (1994)

- If **cointegrated**: incarceration genuinely suppresses crime in the long run
- If **not**: both just trended together — spurious correlation
- Found cointegration **initially**
- Relationship **broke down** in the 1990s when crime fell faster than incarceration rose

Key insight

Cointegration can **falsify** a theory, not just confirm it.
A broken long-run relationship is as informative as a stable one.

Example 4: Democracy & Economic Development

The famous claim (modernisation theory)

Economic growth **causes** democratisation. Both GDP and democracy scores are $I(1)$ and trending upward $\Rightarrow$ they *look* cointegrated.

Acemoglu et al. (2008) — *Journal of Political Economy*

- Once country **fixed effects** removed and cointegration properly tested in panels
- The long-run relationship **largely disappears**
- Both series share a common trend only because both happened to rise in the 20th century

The lesson

Spurious regression was masquerading as a landmark finding in political science.

Cointegration was the tool that **caught it**.

Social Science Parallel: Public Mood & Policy

The data

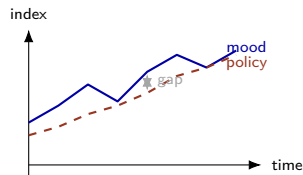
Mood	US Public Mood Index (liberal/conservative composite)
Policy	Policy liberalism index (legislation, spending, rights)
Period	Annual, 1956 – 1996
Source	Erikson, MacKuen & Stimson (2002) <i>The Macro Polity</i>

Social science intuition:

Both series are driven by the same underlying political environment.

Democratic responsiveness predicts:
 $\text{policy} \approx \text{long-run function of public preferences.}$

⇒ They **should** be cointegrated
 with cointegrating vector $\approx [1, -1]$
 (the **opinion–policy gap** is stationary)



Opinion–policy gap = error
 correction term if cointegrated
 Adjustment $\approx 2\text{--}3$ years

What Is the ECM Term? Plain English

One sentence

The ECM term is the **gap between where the system is and where the long-run relationship says it should be.**

Step 1 — find the long-run relationship:

$$\widehat{\text{Policy}}_t = \alpha + \beta \cdot \text{Mood}_t$$

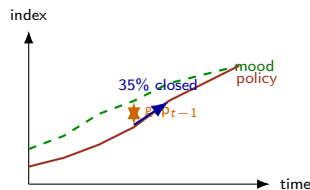
Step 2 — compute the gap (residual):

$$\underbrace{\text{gap}_t = \text{Policy}_t - \hat{\alpha} - \hat{\beta} \cdot \text{Mood}_t}_{\text{ECM term}}$$

Step 3 — the ECM uses it:

$$\Delta \text{Policy}_t = \beta_1 \Delta \text{Mood}_t + \underbrace{\beta_2 \cdot \text{gap}_{t-1}}_{\text{correction}} + \varepsilon_t$$

$\beta_2 < 0$ means the system moves
against the gap — closing it.



gap = political pressure
 β_2 = speed of response
 $1/|\beta_2|$ = years to full correction

ECM in Action: A Numerical Example

Scenario: policy is too conservative

Suppose $\hat{\beta}_2 = -0.35$ (35% of gap closes each year)

Year	Mood	Policy	Gap	What happens next
1	60	45	-15	gap is large → strong pressure
2	61	50	-11	$0.35 \times 15 = 5$ pts corrected
3	61	54	-7	$0.35 \times 11 = 4$ pts corrected
4	62	57	-5	gap shrinking each period
5	62	59	-3	system approaching equilibrium

Key observations

- Gap **shrinks** each year — $\beta_2 < 0$ guarantees this
- Correction is **proportional** — large gap → large correction
- System never overshoots if $-1 < \beta_2 < 0$

Half-life of the gap

How long until half the gap is closed?

$$t_{1/2} = \frac{\ln 0.5}{\ln(1 + \beta_2)}$$

For $\beta_2 = -0.35$:

$$t_{1/2} \approx \frac{-0.693}{-0.421} \approx \mathbf{1.6 \text{ years}}$$

ECM Term: The Three Things to Remember

1

It is the gap
distance between
current state

2

It is the pressure
bigger gap = stronger
force nulling back

3

β_2 is the speed
how much of the gap
closes each period

In politics

Gap = how far policy has drifted
from public opinion.

$\beta_2 = -0.35$ means voters
claw back 35% of drift per year.

In finance

Gap = how far CPI has drifted
from its PPI equilibrium.

$\beta_2 = -0.22$ means 22% of
the price gap closes per month.

Same mathematics. Different story.

The ECM term is always: how wrong is the system right now?

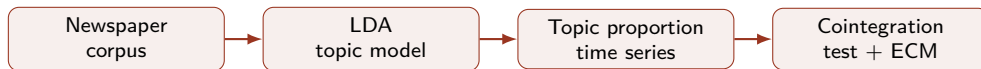
Cutting Edge: Topic Models Meet Cointegration

The idea

Run LDA on a newspaper corpus (e.g. NYT front pages 1980–2020).

Each topic gives a **daily proportion time series**.

These series are often **I(1)** — apply cointegration directly.



Example topics (I(1) series):

- *immigration anxiety*
- *economic crisis*
- *climate change*

Paired with real-world series:

- actual immigration flows
- unemployment rate
- CO₂ measurements

Three Research Questions You Can Answer

1 Media vs. reality: is coverage cointegrated with the phenomenon?

If NYT immigration coverage $\sim I(1)$ and actual flows $\sim I(1)$:

Cointegrated $\Rightarrow$ media tracks reality long-run

Not cointegrated $\Rightarrow$ coverage drifts on its own logic
(*moral panics, editorial cycles*)

The ECM residual = **media distortion** at any point in time

2 Who sets the agenda? Media or the public?

Cointegrate topic coverage with Google Trends for the same topic.

ECM adjustment direction tells you:

Media corrects toward search $\Rightarrow$ **public drives coverage**

Search corrects toward media $\Rightarrow$ **media drives public interest**

3 Do NYT and Fox News share a long-run news agenda?

Cointegrated $\Rightarrow$ common equilibrium, events force convergence

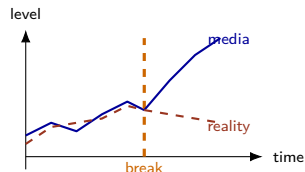
Not cointegrated $\Rightarrow$ **separate information universes**

Granger causality $\Rightarrow$ which outlet *leads*, which *follows*

Structural Breaks: When the Equilibrium Itself Shifts

The most sophisticated application

Test whether the cointegrating relationship **broke down** at a specific date.



The **loss of cointegration** is itself the finding

- Several papers find a structural break **around 2015–2016** in the relationship between economic indicators and media economic coverage
- Before: coverage cointegrated with unemployment, GDP
- After: the long-run equilibrium **disappears**
- Interpretation: media stopped tracking economic reality [Schweikert, 2021, Elejalde et al., 2018, Reuters Institute, 2024]

Granger Causality: The Core Idea

Definition

x_1 **Granger-causes** x_2 if lagged values of x_1 **improve forecasts** of x_2 , beyond what lags of x_2 alone provide.

Formally: estimate two models for x_2 :

Restricted: $x_{2,t} = f(x_{2,t-1}, \dots)$

Unrestricted: $x_{2,t} = f(x_{2,t-1}, \dots, x_{1,t-1}, \dots)$

H_0 : lags of x_1 are **jointly insignificant**

Reject $H_0 \Rightarrow x_1$ Granger-causes x_2

Important caveat

Granger causality is about **predictability**, not **structural causation**.

“ x_1 helps forecast x_2 ”
 $\neq$ “ x_1 causes x_2 ”

Why use first differences?

We run the test on Δ PPI and Δ CPI (stationary series), not levels — OLS on $I(1)$ series gives unreliable F -statistics.

The Drunk and Dog: Granger Version

Who leads, who follows?

The drunk and dog are cointegrated
— but **who is pulling whom?**

- If dog leads: knowing where the dog was yesterday helps predict where the drunk will be today
- If drunk leads: the opposite
- If neither: they just happen to walk together



Dog Granger-causes drunk
(lagged dog position predicts drunk)

Granger causality answers this

It tells you the **direction**
of the long-run relationship,
not just whether one exists.

Granger Test: PPI and CPI

Does PPI Granger-cause CPI?

```
grangertest(x=ppi_cpi$dppi, y=ppi_cpi$dcpi, order=3)
grangertest(x=ppi_cpi$dppi, y=ppi_cpi$dcpi, order=4)
grangertest(x=ppi_cpi$dppi, y=ppi_cpi$dcpi, order=5)
```

H_0 : lags of Δ PPI are insignificant in forecasting Δ CPI

$\Rightarrow$ **Cannot reject** H_0 at 5% $\Rightarrow$ PPI does **not** Granger-cause CPI

Does CPI Granger-cause PPI?

```
grangertest(x=ppi_cpi$dcpi, y=ppi_cpi$dppi, order=5)
```

$\Rightarrow$ **Reject** H_0 at 5% $\Rightarrow$ CPI **does** Granger-cause PPI

Conclusion

CPI $\rightarrow$ PPI

Consumer prices **lead** producer prices.
Retail demand signals transmit
backward through the supply chain.

Bonferroni correction!

Running **multiple** lag orders
inflates Type I error.

Apply Bonferroni: divide
 $\alpha = 0.05$ by number of tests.

How to Test: Engle–Granger Two-Step Procedure

Step 1 — Estimate the cointegrating regression:

```
model.coint <- lm(cpi ~ ppi, data = ppi_cpi)
summary(model.coint)
```

Step 2 — ADF test on the residuals:

```
testdf2(variable = residuals(model.coint),
        test.type = "nc",
        max.augmentations = 4, max.order = 4)
```

- Residuals **stationary** $\Rightarrow$ series are **cointegrated**.
- Residuals **non-stationary** $\Rightarrow$ **not** cointegrated (spurious).

Critical Values: A Special Table Is Required

Important!

You **cannot** use standard ADF critical values on residuals from a cointegrating regression. The residuals are *estimated* — this shifts the distribution of the test statistic.

Sample size ($n \approx 375$)	Critical value (5%)
Engle–Granger critical value	≈ -4.24

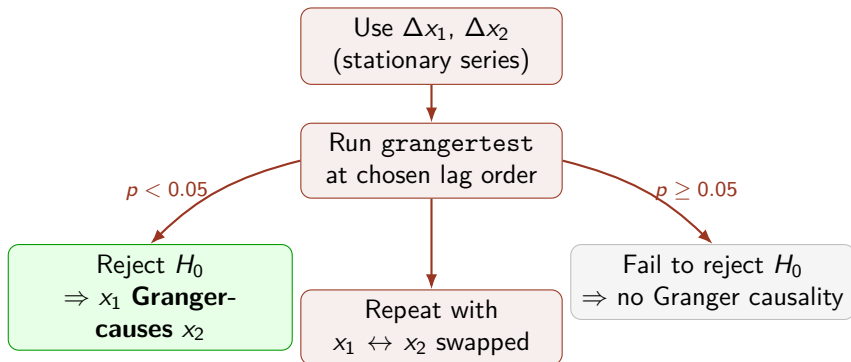
Lab result:

$$\text{Test statistic} = -5.963 < -4.24$$

⇒ Reject H_0 of non-stationarity in residuals.

⇒ CPI and PPI are **cointegrated**. ✓

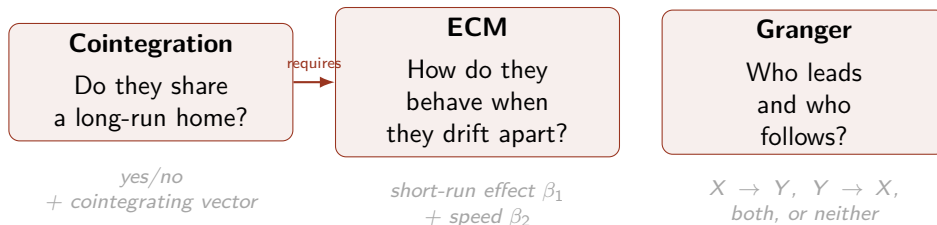
Granger Causality: Summary of Logic



Four possible outcomes

$x_1 \rightarrow x_2$ only	unidirectional causality
$x_2 \rightarrow x_1$ only	unidirectional (reverse)
$x_1 \leftrightarrow x_2$	bidirectional (feedback)
neither	no predictive relationship

Three Different Questions



Key dependency

ECM **requires** cointegration.

No long-run equilibrium

⇒ no ECM term to correct toward.

Granger causality is **independent**

— run it regardless.

They are logically separate

You can have cointegration without knowing the Granger direction.

You can find Granger causality without cointegration.

The Drunk and Dog: All Three at Once

Cointegration

Are the drunk and dog walking to the **same destination**?

They drift apart short-run but **cannot diverge forever**.

⇒ A yes/no property of the system.

ECM

When the dog strays 10 metres, it runs back at **35% per minute**.

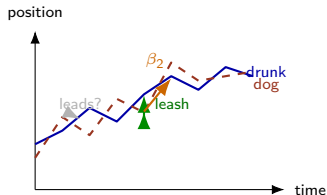
$\beta_2 = -0.35$: the correction mechanism and its speed.

⇒ The dynamic model of return.

Granger causality

Does the dog follow the drunk, or does the drunk follow the dog?

⇒ The direction of navigation.



Green = leash (cointegration)

Orange = correction speed (ECM)

Grey = who navigates (Granger)

Side-by-Side Comparison

	Cointegration	ECM	Granger
Question	Long-run bond?	How to adjust?	Who predicts whom?
Requires	Both $I(1)$	Cointegration	Stationary series
Output	Yes/no + vector	β_1, β_2	Direction
Test	ADF on residuals	OLS on Δ	F -test on lags
If fails	Use Δ only	Cannot estimate	Still informative
Analogy	Same leash?	How fast runs back?	Who navigates?

Thank you